

Installation of Apache Tomcat and Jenkins Pipeline for MyMavenWebApp01

Installation of Apache Tomcat:

Step 1: To install Apache tomcat in root folder, type the command **sudo -i** and then **sudo apt update && sudo upgrade -y**

```
samiksha@samiksha:~$ sudo -i
[sudo] password for samiksha:
root@samiksha:~# sudo apt update && sudo upgrade -y
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
78 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudo: upgrade: command not found
```

Enter tmp by typing the command **cd /tmp** and download Apache Tomcat software by typing the command **wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz**

```
root@samiksha:~# cd /tmp
root@samiksha:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-14 15:46:37-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)[65.108.204.189]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.95.tar.gz'

apache-tomcat-9.0.95.tar.gz 100%[=====>] 12.13M 83.2KB/s in 1m 41s
2026-04-14 15:48:19 (123 KB/s) - 'apache-tomcat-9.0.95.tar.gz' saved [12715996/12715996]
```

Then enter type **cd ..** and extract the downloaded to opt in root by typing the commands

cd opt

mkdir tomcat

cd ..

cd tmp

to extract the tar file into tomcat folder run the command **sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1**

```
root@samiksha:/tmp# cd ..
root@samiksha:/# cd opt
root@samiksha:/opt# cd tmp
-bash: cd: tmp: No such file or directory
root@samiksha:/opt# ls
google gradle
root@samiksha:/opt# mkdir tomcat
root@samiksha:/opt# cd ..
root@samiksha:/# cd tmp
root@samiksha:/tmp# sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1
apache-tomcat-9.0.95/conf/
apache-tomcat-9.0.95/conf/catalina.policy
apache-tomcat-9.0.95/conf/catalina.properties
apache-tomcat-9.0.95/conf/context.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xsd
apache-tomcat-9.0.95/conf/logging.properties
apache-tomcat-9.0.95/conf/server.xml
apache-tomcat-9.0.95/conf/tomcat-users.xml
```

Enter tomcat and type **ls** to see all files are extracted

Enter conf **cd conf** and type **ls** to see server.

```
root@samiksha:/tmp# cd ..
root@samiksha:/# cd opt
root@samiksha:/opt# cd tomcat/
root@samiksha:/opt/tomcat# ls
bin  BUILDING.txt  conf  CONTRIBUTING.md  lib  LICENSE  logs  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  temp  webapps  work
root@samiksha:/opt/tomcat#
root@samiksha:/opt/tomcat# cd conf
root@samiksha:/opt/tomcat/conf# ls
catalina.policy      context.xml          jaspic-providers.xsd  server.xml            tomcat-users.xsd
catalina.properties  jaspic-providers.xml  logging.properties    tomcat-users.xml      web.xml
root@samiksha:/opt/tomcat/conf#
```

xml use this file to change the port number from default 8080 to any number say 9090 because Jenkins is running on 8080

```
root@samiksha:/opt/tomcat/conf# gedit server.xml
```

Type **gedit server.xml** and go to line number 70 and change the portnumber to and save

```
-->
<Connector port="9090" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
    port="8080" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>
```

we need to create tomcat users as well so add the users in tomcat-users.xml file by typing the command **gedit tomcat-users.xml**

```
root@samiksha:/opt/tomcat/conf# gedit tomcat-users.xml
```

go to line 22 add the line **<user username="admin" password="Admin" roles="manager-gui"/>**

```
<tomcat-users xmlns="http://tomcat.apache.org/xml"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
    version="1.0">
    <user username="admin" password="Admin" roles="manager-gui"/>
<!--
```

Then enter bin and type **ls**

```

root@samiksha:/opt/tomcat/conf# cd ..
root@samiksha:/opt/tomcat# cd bin
root@samiksha:/opt/tomcat/bin# ls
bootstrap.jar      ciphers.sh          daemon.sh            setclasspath.bat    startup.sh           version.bat
catalina.bat       commons-daemon.jar  digest.bat           setclasspath.sh     tomcat-juli.jar     version.sh
catalina.sh        commons-daemon-native.tar.gz  digest.sh            shutdown.bat        tomcat-native.tar.gz
catalina-tasks.xml configtest.bat       makebase.bat         shutdown.sh          tool-wrapper.bat
ciphers.bat        configtest.sh       makebase.sh          startup.bat          tool-wrapper.sh

```

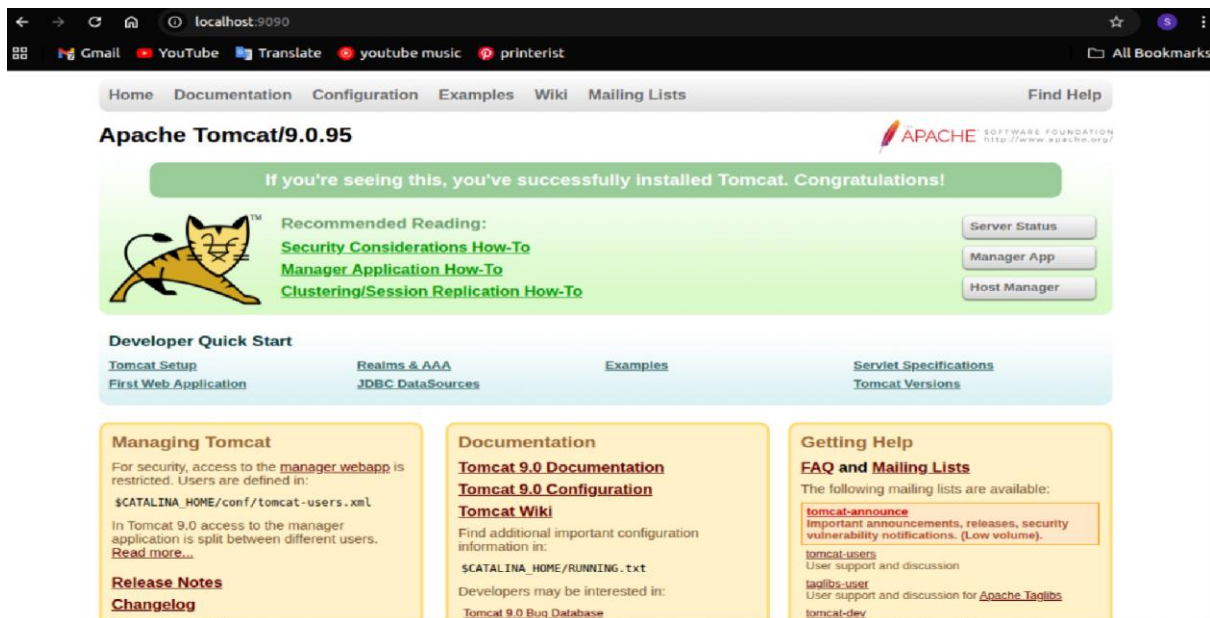
we can type the commands **startup.sh** and **shutdown.sh** to start the tomcat and stop the tomcat

```

root@samiksha:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@samiksha:/opt/tomcat/bin# █

```

open the page in browser say localhost:9090



Click on Manager App enter the credentials previously provided



Tomcat Web Application Manager

Message: OK

Manager
[List Applications](#) [HTML Manager Help](#) [Manager Help](#) [Server Status](#)

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Jenkins Pipeline for MyMavenWebApp01

Step 1: open the terminal and type the command **cd MyMavenWebApp01** and create a **Jenkinsfile** containing the pipeline code, and then commit and push it to GitHub.

```
pipeline {
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/Samiksha201104/MyMavenWebApp01.git'
            }
        }

        stage('Build') {
            steps {
```

```

        sh 'mvn clean package' // Run Maven build
    }}
stage('Test') {
    steps {
        sh 'mvn test' // Run unit tests
    }}
stage('Deploy WAR') {
    steps {
        sh 'cp target/MymavenWebApp01.war /opt/tomcat/webapps/'
    }}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }}
}

```

Go to index.jsp file and edit the hello world code and paste the following code

```

1    <html>
2    <body>
3    <h2>Samiksha P 1BI23CS189</h2>
4    </body>
5    </html>

```

Step 2: Go to Jenkins Dashboard click on new item enter the item name as MyMavenWebApp and select pipeline. In the configure select pipeline scm paste the repository url select github credentials apply and save



New Item

Enter an item name

MyMavenWebApp

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project



Configure

General

Triggers

Pipeline

Advanced

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Samiksha201104/MyMavenWebApp01.git

Credentials ?

GitHub Credentials

+ Add

Advanced

Step 3: go **MyMavenWebApp** in Jenkins and click **Build Now**; the build completes successfully.

MyMavenWebApp - Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Deploy WAR	Declarative: Post Actions
Average stage times: (full run time: ~2min 30s)	3s	652ms	2s	54s	9s	2s	493ms
#1 Apr 15 08:51 No Changes	3s	652ms	2s	54s	9s	2s	493ms

Step 4: go to Tomcat Web Application Manager and select the path /MyMavenWebApp01

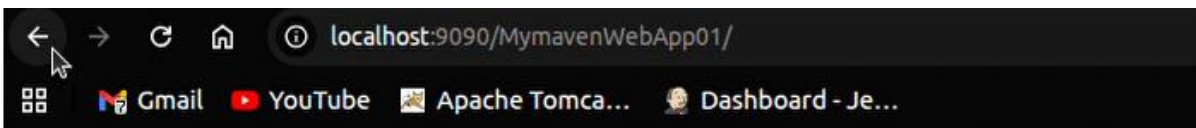


Tomcat Web Application Manager

Message: OK

Manager			
List Applications	HTML Manager Help	Manager Help	Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MymavenWebApp01	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/test-manager	None specified	Tomcat Test Manager Application	true	0	Start Stop Reload Undeploy



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